

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	6	7	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

- Let $\hat{Y}_i = \hat{\theta}_1 x_{i1} + \hat{\theta}_2 x_{i2} + \epsilon_i$ be our regression curve, with $\hat{\theta}_1 \rightarrow$ LSE of θ_1 , $\hat{\theta}_2 \rightarrow$ LSE of θ_2

$$i) \hat{Y}_1 = \hat{\theta}_1 + \hat{\theta}_2 + \epsilon_1$$

$$ii) \hat{Y}_2 = \hat{\theta}_1 - 2\hat{\theta}_2 + \epsilon_2$$

$$iii) \hat{Y}_3 = 2\hat{\theta}_1 - \hat{\theta}_2 + \epsilon_3$$

• ~~Loss~~ Residual = $Q = (y_1 - \hat{Y}_1)^2 + (y_2 - \hat{Y}_2)^2 + (y_3 - \hat{Y}_3)^2$

$$= (y_1 - \hat{\theta}_1 - \hat{\theta}_2)^2 + (y_2 - \hat{\theta}_1 + 2\hat{\theta}_2)^2 + (y_3 - 2\hat{\theta}_1 + \hat{\theta}_2)^2 = 0$$

• To minimize residual:

• Now $\frac{\partial Q}{\partial \hat{\theta}_1} = 0$ and $\frac{\partial Q}{\partial \hat{\theta}_2} = 0$

$$i) \frac{\partial Q}{\partial \hat{\theta}_1} = 0 \Rightarrow -2(y_1 - \hat{\theta}_1 - \hat{\theta}_2) - 2(y_2 - \hat{\theta}_1 + 2\hat{\theta}_2) - 4(y_3 - 2\hat{\theta}_1 + \hat{\theta}_2) = 0$$

$$\Rightarrow -2y_1 - 2y_2 - 4y_3 + \hat{\theta}_1(2+2+8) + \hat{\theta}_2(2-4-4) = 0$$

$$\Rightarrow -2y_1 - 2y_2 - 4y_3 + 12\hat{\theta}_1 - 6\hat{\theta}_2 = 0 \quad \text{--- (1)}$$

$$ii) \frac{\partial Q}{\partial \hat{\theta}_2} = 0 \Rightarrow -2(y_1 - \hat{\theta}_1 - \hat{\theta}_2) + 4(y_2 - \hat{\theta}_1 + 2\hat{\theta}_2) + 2(y_3 - 2\hat{\theta}_1 + \hat{\theta}_2) = 0$$

$$\Rightarrow -2y_1 + 4y_2 + 2y_3 + \hat{\theta}_1(2-4-4) + \hat{\theta}_2(2+8+2) = 0$$

$$\Rightarrow -2y_1 + 4y_2 + 2y_3 - 6\hat{\theta}_1 + 12\hat{\theta}_2 = 0 \quad \text{--- (2)}$$

$$(1) + 2 \times (2) \Rightarrow -6y_1 + 6y_2 + 18\hat{\theta}_2 = 0 \Rightarrow \hat{\theta}_2 = \frac{6y_2 - 6y_1}{18}$$

$$\Rightarrow \hat{\theta}_2 = \frac{y_2 - y_1}{3}$$

~~$\Rightarrow$ (1) bpo~~ Now (2) + 2x(1)

$$\Rightarrow -6y_1 - 6y_3 + 18\hat{\theta}_1 = 0$$

$$\Rightarrow \hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\therefore \text{LSE of } \theta_1 = \hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\text{LSE of } \theta_2 = \hat{\theta}_2 = \frac{y_2 - y_1}{3}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

a) The ~~correlation~~ ^{var-covariance} matrix of $\mathbf{X}$ would look like this:

$$\begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \text{Cov}(X_1, X_3) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \text{Cov}(X_2, X_3) \\ \text{Cov}(X_3, X_1) & \text{Cov}(X_3, X_2) & \text{Var}(X_3) \end{bmatrix} \text{ after standardization}$$

- $\text{Var}(X_1) = \text{Var}(U_1) \neq$

- We know that variance of a uniform random variable on interval (a, b) is $\frac{(b-a)^2}{12}$

$$\therefore \text{Var}(X_1) = \text{Var}(U_1) = \frac{(1-0)^2}{12} = \frac{1}{12}$$

$$\text{Similarly } \text{Var}(X_2) = \text{Var}(U_2) = \frac{1}{12}$$

$$\text{Var}(X_3) = \text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2)$$

Since U_1 & U_2 are independent $\Rightarrow \text{Cov}(U_1, U_2) = 0$

$$\Rightarrow \text{Var}(X_3) = \text{Var}(U_1) + \text{Var}(U_2) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

- Now covariances: $\text{Cov}(X_1, X_2) = \text{Cov}(U_1, U_2) = 0$

$$\text{Similarly } \text{Cov}(X_2, X_1) = \text{Cov}(U_2, U_1) = 0$$

- $\text{Cov}(X_1, X_3) = \text{Cov}(U_1, U_1 + U_2) = \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2)$
 $= \text{Var}(U_1) + 0 = \frac{1}{12} = \text{Cov}(X_3, X_1)$

- $\text{Cov}(X_2, X_3) = \text{Cov}(U_2, U_1 + U_2) = \text{Cov}(U_2, U_1) + \text{Cov}(U_2, U_2)$
 $= 0 + \text{Var}(U_2) = \frac{1}{12}$
 $= \text{Cov}(X_3, X_2)$

∴ Var - covariance matrix of X :

$$\begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{2}{12} \end{bmatrix}$$

- After normalization: Correlation matrix of $X = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} = \rho$
- Using: $\text{corr}(X_i, X_j) = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i)} \sqrt{\text{Var}(X_j)}}$

b) To find first The variance of i^{th} principal component is the value of the i^{th} eigenvalue after arranging in descending order.

- Since we need var. of 1st principal component, we need biggest eigenvalue of ρ .

• $\det[\rho - \lambda I] = 0$ for eigenvalue vector λ

$$\Rightarrow \left| \begin{bmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{bmatrix} \right| = 0$$

$$\Rightarrow [(1-\lambda)[(1-\lambda)^2 - \frac{1}{2}] + \frac{1}{\sqrt{2}} [-(1-\lambda)(\frac{1}{\sqrt{2}})] = 0$$

$$\Rightarrow (1-\lambda)^3 - \frac{(1-\lambda)}{2} - \frac{(1-\lambda)}{2} = 0$$

$$\Rightarrow (1-\lambda)^3 - (1-\lambda) = 0$$

$$\Rightarrow (1-\lambda)[(1-\lambda)^2 - 1] = 0$$

$$\Rightarrow (1-\lambda)[\lambda^2 - 2\lambda] = 0$$

$$\Rightarrow (1-\lambda)\lambda(\lambda-2) = 0$$

$$\Rightarrow \lambda_1 = +1, \lambda_2 = 0, \lambda_3 = 2 \text{ are the zeroes.}$$

∴ Our required value is 2 because it is the greatest.

∴ Variance of 1st principal component is 2.

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) complete $\Rightarrow$ Mon

- Start with lowest distance $\rightarrow 1 \rightarrow$ betⁿ 1 & 2
 $\Rightarrow$ cluster $\{1, 2\}$

	$\{1, 2\}$	3	4
$\{1, 2\}$	0		
3	11	0	
4	5	3.5	0

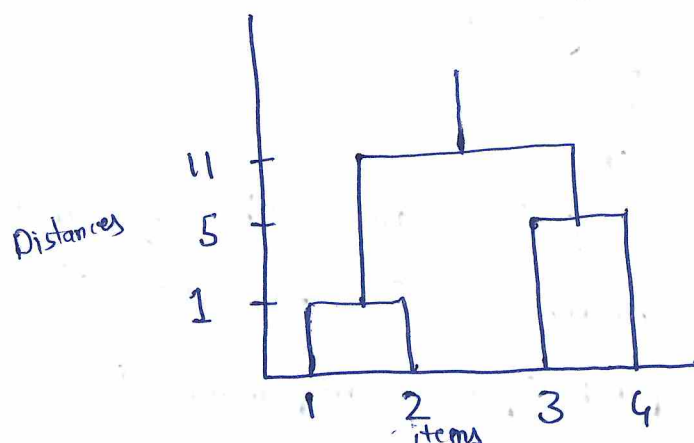
$\rightarrow$ Next min^m distance is betⁿ 3 & 4 $\Rightarrow$ cluster them

	$\{1, 2\}$	$\{3, 4\}$
$\{1, 2\}$	0	
$\{3, 4\}$	11	0

How?

$\rightarrow$ last we cluster $\{1, 2\}$ & $\{3, 4\}$ into $\{1, 2, 3, 4\}$

b)
Dendrogram:



a) ii) Average linkage

- Start with lowest distance $\rightarrow 1 \rightarrow \text{bet}^n 1 \text{ \& } 2$
 $\Rightarrow$ cluster $\{1, 2\}$

$$\begin{array}{c} \{1,2\} \quad 3 \quad 4 \\ \begin{array}{c} \{1,2\} \\ 3 \\ 4 \end{array} \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{array}$$

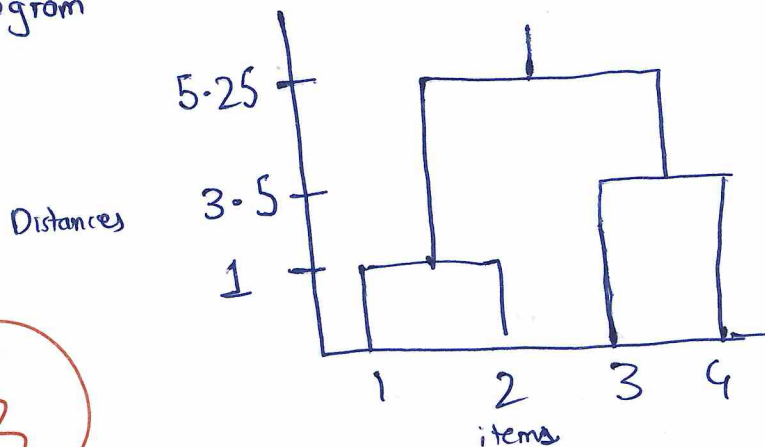
How??

- Next min^m dist is $3.5 \rightarrow \text{bet}^n 3 \text{ \& } 4 \Rightarrow$ cluster them

$$\begin{array}{c} \{1,2\} \quad \{3,4\} \\ \begin{array}{c} \{1,2\} \\ \{3,4\} \end{array} \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{array}$$

- Next we cluster $\{1,2\}$ & $\{3,4\}$ into $\{1,2,3,4\}$

b) Dendrogram



On comparing these two, we find that the distances at which clustering is made is lower in case (ii) because we took averages instead of maximums

- Still, the clusters formed at each step remains same.

ADDITIONAL PAGES FOR ANSWERS

